SIMATS ENGINEERING
CSA6303- Threat Intelligence and Network Security
CO3-AT2-Malware Analysis Assignment

Assessment Questions

Subject	Threat Intelligence	**Total Marks**	50
Questions	10 (all compulsory)	**Marks per Question**	5
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Instructions: Answer all questions. Each question carries 5 marks. Support answers with examples and diagrams where appropriate.

CO3: Perform threat analysis using intelligence models and utilize industry tools for monitoring and detection.

Code of Conduct:

I B.Hari Sai/192311098 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of Student:B.Hari Sai

Questions :

1. Malware Analysis Using Static Techniques:A security analyst receives a suspicious executable file attached to a phishing email. Before running the file, the analyst wants to examine its properties, file hash, strings, and imported functions. Explain how static malware analysis can help identify suspicious characteristics and assess the potential risk without executing the file.

2. Malware Analysis Using Dynamic Techniques: An organization suspects that an employee has downloaded a malicious program that may modify system files and communicate with an external server. Explain how dynamic malware analysis can be performed in an isolated virtual environment to observe the program's behavior and identify potential Indicators of Compromise (IoCs).

3. Malware Investigation Using a Threat Intelligence Platform: A company's security team discovers a suspicious file on an employee's computer. The file hash appears in a threat intelligence platform associated with a known malware family. Explain how the analyst can use threat intelligence sources to investigate the file, assess the reliability of the information, and recommend suitable containment measures.

4. Malware Detection Through Data Collection and Correlation: A university detects unusual processes on several computers, suspicious DNS requests, and unexpected outbound network connections. Explain how an analyst can collect and correlate endpoint logs, DNS records, firewall logs, and file hashes to determine whether the computers are infected with malware.

5. Malware Investigation Using SIEM Tools:A financial institution's Security Information and Event Management (SIEM) system generates alerts about suspicious process execution, repeated failed logins, and unusual network connections from the same workstation. Describe how the security analyst can use SIEM capabilities to correlate these events, investigate the possible malware infection, and prioritize the incident.

6. Malware Detection and Log Analysis Using Splunk: A company uses Splunk to monitor its network and endpoint logs. The security team suspects that malware is attempting to contact an external command-and-control server. Explain how Splunk can be used to search relevant logs, identify suspicious communication patterns, correlate events, and document evidence of possible malware activity.

7. Malware Investigation Using the ELK Stack: An e-commerce company uses the ELK Stack (Elasticsearch, Logstash, and Kibana) to monitor web server logs, firewall records, and endpoint events. The security team notices repeated connections to suspicious domains and unusual file activity. Explain how the ELK Stack can support malware investigation through log collection, processing, searching, visualization, and correlation.

8. Threat Hunting for Malware Activities: An organization's security team suspects that malware has infected an employee workstation but has not received any antivirus alerts. The workstation shows unusual CPU usage, unfamiliar processes, and connections to unknown external IP addresses. Describe a threat-hunting approach to investigate the workstation, identify suspicious behaviors, collect relevant IoCs, and recommend remediation measures.

9. Comparing Malware Analysis Tools Used in the IT Industry:A cybersecurity team needs to investigate a suspicious file and determine whether it contains malicious code. Compare static analysis tools, dynamic analysis tools, network traffic analyzers, and threat intelligence platforms in terms of their purpose and usefulness. Recommend an appropriate combination of tools for conducting a malware investigation.

10. Malware Analysis Report and Incident Response: A company discovers a suspicious executable on an employee's computer. Initial analysis identifies an unfamiliar process, a modified system configuration, and communication with a suspicious external domain. Prepare an outline for a malware analysis report that includes file information, analysis methodology, observed behavior, IoCs, threat intelligence findings, risk assessment, and recommended containment and recovery actions.

ANSWERS:

1. Malware Analysis Using Static Techniques

Static malware analysis is the process of examining a suspicious file without executing it. It allows a security analyst to identify potentially harmful characteristics while reducing the risk of infecting a computer. The analyst first records basic file properties, including the file name, size, type, location, and timestamps. A cryptographic hash, such as SHA-256, is then calculated to create a unique fingerprint that can be checked against malware databases and threat-intelligence platforms.

The analyst next uses file-identification tools, string extraction utilities, PE viewers, and disassemblers to examine the file's internal structure. Readable strings may reveal suspicious URLs, IP addresses, commands, file paths, registry keys, or messages. Imported functions can also provide clues: functions associated with process creation, file modification, registry changes, encryption, or network communication may indicate capabilities that require further investigation. Unusual file sections, packing, obfuscation, or a mismatch between the file extension and its actual format may also be suspicious.

Static analysis is valuable because it is relatively safe and can quickly provide useful information before the sample is run. However, malware may hide its code through encryption or obfuscation, and some behavior only appears during execution. Therefore, the analyst should treat these findings as evidence for further investigation rather than relying on one suspicious characteristic alone.

2. Malware Analysis Using Dynamic Techniques

Dynamic malware analysis involves executing a suspicious program in a controlled environment and observing what it does. The analyst should prepare an isolated virtual machine with a clean snapshot, no personal or organizational data, and restricted networking. Shared folders, clipboard sharing, and access to the host should be disabled where possible. Before running the sample, the analyst records the baseline state of the machine, including running processes, files, registry settings, and network activity.

After execution, monitoring tools are used to observe process creation, parent-child process relationships, file creation or deletion, registry modifications, startup changes, scheduled tasks, and attempts to gain persistence. Network monitoring tools record DNS lookups, destination IP addresses, ports, protocols, and requests to external servers. The analyst compares these observations with the baseline and records the time at which each action occurs. This can show whether a program modifies system files or attempts to contact a command-and-control server.

Potential Indicators of Compromise (IoCs) include file hashes, dropped filenames, suspicious process paths, registry keys, domains, IP addresses, and unusual network patterns. The analyst should preserve relevant logs and packet captures, then revert the virtual machine to its clean snapshot. Dynamic analysis can reveal behavior that static analysis misses, but malware may detect a virtual environment or delay its actions. For this reason, the results should be combined with static analysis and threat intelligence.

3. Malware Investigation Using a Threat Intelligence Platform

When a suspicious file is discovered, the analyst calculates its SHA-256 hash and searches for it in a threat-intelligence platform. The platform may provide a malware-family classification, antivirus detection results, related hashes, known domains and IP addresses, behavioral reports, first-seen dates,

and information from earlier investigations. This information can help the analyst understand whether the file is already known and what activities it may perform.

The reliability of the results must be evaluated carefully. The analyst should consider the reputation of the source, the date of the report, the evidence supporting the classification, the confidence level, and whether several independent sources agree. A single detection label or an old community report may be inaccurate or incomplete. The analyst should compare the intelligence findings with the file's properties, observed behavior, and local endpoint logs before deciding that the file is malicious.

If the evidence indicates a likely infection, the affected endpoint should be isolated from the network, relevant logs and the sample should be preserved, and confirmed malicious domains or IP addresses may be blocked through approved security controls. The security team should search other systems for matching hashes and related IoCs. After determining the scope, the team can remove the malware, eliminate persistence mechanisms, patch the weakness used for entry, and restore the system if necessary. Threat intelligence speeds up investigation, but containment decisions should be based on corroborated evidence and the potential business impact.

4. Malware Detection Through Data Collection and Correlation

To investigate possible infections across several university computers, the analyst collects endpoint logs, process events, DNS records, firewall logs, network-flow information, and file hashes. Each record should include useful fields such as timestamp, hostname, user account, process name, process ID, source and destination addresses, domain, and event type. Time synchronization is important because records from different systems must be compared accurately.

Endpoint logs can reveal unfamiliar processes, unusual command lines, recently created files, or changes to startup settings. DNS records may show repeated lookups for suspicious or newly observed domains, while firewall logs can indicate whether the computer successfully connected to a related external IP address. File hashes can be checked against threat-intelligence sources and compared across computers to determine whether the same suspicious file exists elsewhere.

The analyst correlates the information into a timeline. For example, a file may be downloaded, a new process may start shortly afterward, a DNS request may follow, and the firewall may record an outbound connection to the resolved address. When these events occur together and have no legitimate explanation, they provide stronger evidence of malware activity than any single log entry. The analyst should validate the pattern against normal software, identify affected systems, preserve the evidence, and isolate confirmed or highly suspected computers.

5. Malware Investigation Using SIEM Tools

A Security Information and Event Management (SIEM) system collects and centralizes security events from endpoints, servers, identity systems, and network devices. When alerts indicate suspicious process execution, repeated failed logins, and unusual network connections from the same workstation, the analyst begins by reviewing the alert details, affected host, user account, timestamps, severity, and related events.

SIEM searches and correlation rules help determine whether the events are connected. For example, several failed login attempts followed by a successful login, execution of an unfamiliar program, and an outbound connection to a rare external address may indicate account compromise followed by malware execution. The analyst examines command-line arguments, process ancestry, destination reputation, login source, and whether similar events occurred on other systems. A timeline helps clarify the sequence and possible cause of the activity.

The incident should be prioritized according to confidence, system criticality, user privileges, data sensitivity, possibility of spread, and evidence of active command-and-control communication. A suspected infection on a critical financial system or one involving active data transfer deserves urgent attention. The analyst preserves logs, escalates the case according to policy, isolates the endpoint when authorized, blocks verified indicators, and records the evidence and response actions in the SIEM case-management system. The final investigation should also identify detection gaps and improvements.

6. Malware Detection and Log Analysis Using Splunk

Splunk can collect and search logs from endpoint protection systems, operating systems, DNS servers, proxies, and firewalls. To investigate suspected command-and-control activity, the analyst first selects the relevant time range and filters events for the affected workstation. Searches can look for repeated DNS requests, connections to rare external IP addresses, unusual ports, suspicious process names, and command lines that launch scripts or tools unexpectedly.

A simple illustrative Splunk search is shown below. The field names and index should be adjusted to match the organization's data sources:

```
index=* host="WORKSTATION-NAME" (dest_ip=* OR query=*)
| stats count values(dest_ip) values(query) by host, process_name
```

The analyst reviews the returned events and compares their timestamps. If a suspicious process starts shortly before repeated DNS queries and outbound connections, the sequence may indicate that the process is communicating with a remote server. Searches across multiple hosts can show whether the same domain, IP address, or file hash appears elsewhere. Dashboards and time-based visualizations can help identify repeated beacon-like connections or sudden increases in traffic.

Splunk findings should be validated against the original source logs and reputable threat-intelligence information. The analyst saves relevant events, documents search criteria and timestamps, builds an incident timeline, and records possible IoCs. Detection rules can then be created or improved to alert on similar patterns in the future. Unfamiliar traffic alone should not be blocked automatically without checking whether it belongs to legitimate business software.

7. Malware Investigation Using the ELK Stack

The ELK Stack—Elasticsearch, Logstash, and Kibana—supports malware investigations by collecting, processing, indexing, searching, and visualizing security events. Logstash receives records from web servers, firewalls, and endpoint agents. It parses and standardizes fields such as timestamps, hostnames, process names, domains, destination IP addresses, and file paths. Elasticsearch stores the structured events so they can be searched efficiently, while Kibana provides the interface for filtering and visualization.

The analyst uses Kibana to investigate repeated connections to suspicious domains and unusual file activity. Events can be filtered by host, time range, process, domain, IP address, or filename. Visualizations may reveal sudden increases in outbound connections, repeated requests to the same destination, new files created shortly before network activity, or similar events occurring on multiple computers. Correlating endpoint and firewall records can help establish whether a process initiated a connection to a suspicious server.

The resulting timeline may show a suspicious process starting, a DNS lookup occurring, and a connection being recorded by the firewall. The analyst compares the indicators with reliable threat-intelligence sources, validates findings against original logs, and records hashes, domains, IP addresses, file paths, and timestamps. Kibana dashboards and detection rules can support future monitoring.

Accurate parsing and synchronized timestamps are essential, because incorrectly processed fields or inconsistent times can lead to misleading conclusions.

8. Threat Hunting for Malware Activities

Threat hunting is a proactive investigation intended to find malicious activity that automated security tools may have missed. The analyst begins with a hypothesis: malware may be running on the workstation despite the absence of an antivirus alert. The investigation collects information about running processes, parent-child process relationships, command lines, CPU and memory usage, recently changed files, startup entries, scheduled tasks, and network connections.

The analyst looks for suspicious signs such as unfamiliar or unsigned executables, programs running from temporary or user-writable folders, unexpected scripting activity, unusual persistence methods, and connections to rare or unrecognized external IP addresses. These observations are correlated with DNS, firewall, authentication, and endpoint-protection logs. The analyst calculates hashes for suspicious files and records domains, IP addresses, filenames, process paths, registry keys, and relevant timestamps as potential IoCs.

Each finding should be checked against approved applications and normal administrative activity to reduce false positives. If the evidence indicates compromise, the workstation should be isolated, forensic evidence preserved, and confirmed indicators blocked. The team should remove persistence, patch the exploited weakness, and restore the system from a trusted backup or reimage it if its integrity cannot be guaranteed. Finally, the same IoCs should be searched across the wider organization, and detection rules should be improved using the lessons learned.

9. Comparing Malware Analysis Tools Used in the IT Industry

Static analysis tools examine a file without executing it. Hash utilities, file-identification tools, strings, PE viewers, and disassemblers can reveal file structure, embedded text, imported functions, and possible capabilities. They are useful for initial triage and are comparatively safe, but packing and obfuscation can hide important details.

Dynamic analysis tools run a suspicious sample inside an isolated sandbox or virtual machine and monitor its runtime behavior. They can reveal process creation, file and registry changes, persistence attempts, and network communication. However, they require careful isolation, and a sample may detect the analysis environment or delay its behavior. Network traffic analyzers such as Wireshark inspect packets, DNS requests, protocols, and destination addresses, helping the analyst understand how the sample communicates. Threat-intelligence platforms provide reputation data, malware associations, related indicators, and reports from previous investigations, although the quality and freshness of their information can vary.

A practical investigation should combine these tools rather than rely on only one. First, calculate the file hash and inspect the file type, strings, and imports. Next, if required, execute the sample in a carefully isolated sandbox while collecting process and network activity. Use a network analyzer to examine the traffic and compare the resulting hashes, domains, and IP addresses with several reputable intelligence sources. Finally, correlate all findings, distinguish confirmed facts from assumptions, and document the confidence and limitations of the conclusion. This combination provides a more complete assessment of the file and its potential impact.

10. Malware Analysis Report and Incident Response

A malware analysis report provides a structured and reproducible record of what was discovered, how the investigation was performed, what evidence was observed, and what actions are needed. It helps technical teams, managers, and incident responders understand the risk and coordinate their response. A suitable report can contain the following sections:

1. Executive summary: Briefly describe the discovery, suspected threat, affected workstation, potential impact, and current status.
2. File information: Record the filename, file type, size, storage path, timestamps, and SHA-256 hash. Preserve the original sample securely and document how it was handled.
3. Analysis methodology: Explain the static analysis, sandbox or dynamic analysis, log review, network inspection, and threat-intelligence sources used. Include the tools and relevant time periods.
4. Observed behavior: Describe the unfamiliar process, modified system configuration, files created or changed, persistence attempts, child processes, and communications with external domains or IP addresses.
5. Indicators of Compromise: List hashes, filenames, process paths, domains, IP addresses, URLs, registry keys, and timestamps. Include context and confidence for each indicator.
6. Threat intelligence findings: Summarize related reports, possible malware-family classification, source reliability, report dates, and any conflicting information.
7. Risk assessment: Explain potential effects on confidentiality, integrity, and availability; the importance of affected assets; the likelihood of spread; and the confidence in the assessment.
8. Containment and recovery: Isolate the affected workstation, block verified malicious indicators, preserve evidence, inspect other systems, remove persistence, patch weaknesses, and restore from trusted backups or reimage when necessary.
9. Validation and lessons learned: Confirm that suspicious activity has stopped, monitor for recurrence, document the incident timeline and decisions, and recommend improvements to security controls and detection rules.

The report should separate confirmed observations from assumptions and clearly identify any limitations in the analysis. Evidence should be stored securely with timestamps and traceable handling records. After recovery, the security team should continue monitoring for related activity and share relevant findings with authorized stakeholders so that similar incidents can be detected and prevented earlier.